Project Progress Report — Week 1: Problem Definition & Literature Survey

Work done in current Week:

- Researched UPI fraud vectors focusing on mule accounts, phishing, and rapid layering.
- Finalized problem statement, literature survey, and project requirement specifications.

Work to be done in next Week :

- Study Graph Neural Network algorithms (GCN vs GraphSAGE) and design entity schema.

Supervisor Comments:

Good literature review. Problem statement clearly identifies relational graph advantages.

Supervisor Signature/Date: .....

Project Progress Report — Week 2: Synthetic Data Generation

Work done in current Week:

- Developed synthetic UPI transaction generator (100,000 transactions across 5,000 users).
- Injected 5 realistic fraud patterns (mule rings, nocturnal transfers, velocity bursts).

Work to be done in next Week :

- Perform data preprocessing, cyclic time encoding, and feature normalization.

Supervisor Comments:

Realistic data distribution. Ensure 1.5% fraud class imbalance is maintained.

Supervisor Signature/Date: .....

Project Progress Report — Week 3: Feature Engineering & Graph Construction

Work done in current Week:

- Built interaction graph connecting transactions via shared Device IDs and proxy IPs.
- Applied degree capping on high-volume merchant nodes to prevent over-smoothing.

Work to be done in next Week :

- Split graph into train, validation, and test masks using temporal ordering.

Supervisor Comments:

Graph construction is sound. Temporal ordering prevents data leakage.

Supervisor Signature/Date: .....

Project Progress Report — Week 4: GraphSAGE Architecture Design

Work done in current Week:

- Implemented 2-layer Inductive GraphSAGE model in PyTorch Geometric.
- Constructed baseline models (XGBoost and 3-layer MLP) for empirical benchmarking.

Work to be done in next Week :

- Formulate custom loss function to address severe 1.5% class imbalance.

Supervisor Comments:

Inductive GraphSAGE is appropriate for dynamic real-time user onboarding.

Supervisor Signature/Date: .....

Project Progress Report — Week 5: Loss Function (Sigmoid Focal Loss)

Work done in current Week:

- Implemented Sigmoid Focal Loss (alpha=0.85, gamma=2.0) to penalize easy benign samples.
- Configured Adam optimizer (lr=0.003) and ReduceLROnPlateau learning rate scheduler.

Work to be done in next Week :

- Train GraphSAGE and perform validation threshold tuning on PR-curve.

Supervisor Comments:

Mathematical formulation of Focal Loss is sound. Monitor validation loss curves.

Supervisor Signature/Date: .....

Project Progress Report — Week 6: Model Training & Baseline Benchmarking

Work done in current Week:

- Trained GraphSAGE over 35 epochs; achieved 99.62% precision with 0.884 PR-AUC.
- Evaluated test set: only 1 false positive out of 14,727 transactions (beating XGBoost by 14%).

Work to be done in next Week :

- Build deterministic rule engine for cold-start and instant velocity checks.

Supervisor Comments:

Excellent benchmark results. High precision preserves clean customer experience.

Supervisor Signature/Date: .....

Project Progress Report — Week 7: 70/30 Hybrid Decision Engine

Work done in current Week:

- Formulated Hybrid Engine: Final Risk = 0.70 * GNN Score + 0.30 * Rule Score.
- Added Explainable AI (XAI) audit tagging for banking and RBI regulatory compliance.

Work to be done in next Week :

- Build production asynchronous FastAPI backend microservice.

Supervisor Comments:

Hybrid architecture effectively solves cold-start and regulatory explainability.

Supervisor Signature/Date: .....

Project Progress Report — Week 8: FastAPI Backend & Latency Benchmarks

Work done in current Week:

- Developed high-throughput FastAPI service with /predict, /stats, and /health endpoints.
- Profiled single-sample latency: achieved 6.4 ms (well within NPCI's 50 ms SLA).

Work to be done in next Week :

- Build interactive frontend dashboard with live network topology visualization.

Supervisor Comments:

Sub-10ms latency demonstrates production-grade real-time capability.

Supervisor Signature/Date: .....

Project Progress Report — Week 9: Wise-Themed Dashboard & Vis.js Topology

Work done in current Week:

- Designed frontend in Wise Design System with centered Vis.js ForceAtlas2 network graph.
- Fixed graph spring jittering using 0.92 damping, minVelocity cutoff, and radial anchors.

Work to be done in next Week :

- Deploy multi-service project on Vercel and execute test suites.

Supervisor Comments:

Visual interface is exceptionally clean. Mule ring simulation works smoothly.

Supervisor Signature/Date: .....

Project Progress Report — Week 10: Cloud Deployment & Final Verification

Work done in current Week:

- Configured Vercel multi-service deployment with decoupled edge frontend and serverless API.
- Verified 10/10 automated pytest unit suites and completed thesis viva preparation.

Work to be done in next Week :

- Final academic presentation, live demonstration, and project thesis submission.

Supervisor Comments:

All project milestones completed successfully. Approved for final evaluation.

Supervisor Signature/Date: .....

Project Progress Report — Week 11: Adversarial Stress Testing & Edge Cases

Work done in current Week:

- Conducted adversarial stress testing against Sybil attacks and rapid device ID spoofing.
- Verified that Rule Engine catches borderline cases where GNN confidence dips below 0.50.

Work to be done in next Week :

- Integrate GNNExplainer for subgraph visualization and nodal feature importance.

Supervisor Comments:

Adversarial evaluation confirms robustness against evasion attempts by fraud syndicates.

Supervisor Signature/Date: .....

Project Progress Report — Week 12: Explainable AI (XAI) & Interpretability

Work done in current Week:

- Implemented GNNExplainer integration to extract 2-hop topological attribution subgraphs.
- Mapped model decisions to human-readable compliance tags for banking and RBI audit trails.

Work to be done in next Week :

- Execute multi-threaded concurrency and high-throughput stress testing.

Supervisor Comments:

XAI features provide strong regulatory compliance for financial decision audits.

Supervisor Signature/Date: .....

Project Progress Report — Week 13: High Concurrency & Load Testing

Work done in current Week:

- Benchmarked system under simulated high load of 1,000 transactions/second on multi-core CPU.
- Maintained median P50 latency of 5.8 ms with zero dropped transactions or memory leaks.

Work to be done in next Week :

- Compile comprehensive project thesis, IEEE format research paper, and user guide.

Supervisor Comments:

Throughput and memory stability meet commercial payment processing criteria.

Supervisor Signature/Date: .....

Project Progress Report — Week 14: Research Paper Drafting & Documentation

Work done in current Week:

- Drafted research paper titled 'Real-Time UPI Fraud Detection Using Inductive GraphSAGE'.
- Compiled complete technical thesis chapters covering methodology, loss formulas, and benchmarks.

Work to be done in next Week :

- Conduct mock viva presentations and practice interactive UI demo workflows.

Supervisor Comments:

Documentation and research paper draft are well-structured and technically rigorous.

Supervisor Signature/Date: .....

Project Progress Report — Week 15: Mock Viva Defense & Final Submission

Work done in current Week:

- Rehearsed live examiner demo covering Auto-Stream, clean transactions, and mule ring attack.
- Finalized project deliverables, code repository archiving, and supervisor sign-offs.

Work to be done in next Week :

- Appear for final external project viva evaluation and committee defense.

Supervisor Comments:

Excellent execution across all engineering milestones. Fully approved for final examination.

Supervisor Signature/Date: .....